The Zero-Energy Principle of Flexible Spacetime: A Geometric Duality Theory of Matter and Space

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Abstract

This paper investigates the dynamics of spacetime geometry and matter fields as a unified entity, based on the fundamental hypothesis that spacetime is intrinsically flexible. The study develops from a concrete tachyon-geometric bound state model established in prior work. We first rigorously prove that this bound state satisfies a zero-energy condition, with its total energy precisely vanishing, achieving perfect balance between the negative potential energy of matter and the positive confinement energy of geometry. Further linear stability analysis demonstrates that this configuration is stable under perturbations, with the stability mechanism arising from dynamic feedback regulation where geometric deformations respond to field fluctuations. From this specific solution, we extract its core geometry-matter adaptation coupling principle. By introducing an adaptation functional $C[g,\Phi]$ that characterizes the strength of this coupling, this principle is generalized into a universal theoretical framework. This framework naturally describes the co-evolution dynamics of matter fields and flexible geometry, accommodates Standard Model matter fields, and contains the aforementioned zero-energy bound state as a special solution where $C[g,\Phi]=0$. This research provides a novel theoretical perspective for understanding the deep connection between spacetime and matter, as well as the origin of system stability. This work employs a perturbative, iterative approach, taking the solution in flat spacetime as the zeroth-order approximation, to preliminarily explore the possible morphology and properties of such a 'geometry-matter bound state,' thereby laying the theoretical groundwork for its complete, self-consistent solution

Keywords: *Flexible spacetime; Geometry-matter bound state; Zero-energy condition; Dynamical stability; Adaptation functional; Coupling principle.*

I. Introduction

The nature of space, and its unification with time as spacetime, constitutes a central evolving theme in physics. The Newtonian framework established the concepts of absolute space and time as a fixed, immutable stage [1]. This view was fundamentally revised by Einstein's theories of relativity. Special Relativity fused space and time into a four-dimensional Minkowskian continuum, though this spacetime remained a fixed background [2]. General Relativity completed a profound shift by dynamizing geometry, whereby spacetime curvature is determined by the stress-energy content of matter, establishing a dynamic interplay [3]. The quest for a quantum theory of gravity has since led to a multiplicity of paradigms, each proposing a radically different fundamental nature for spacetime. For instance, Loop Quantum Gravity posits a discrete, granular structure emerging from quantum geometric states [4,5], while String Theory requires extra spatial dimensions compactified on complex manifolds, with our 4D universe as a low-energy description [6,7]. Perhaps the most radical departure comes from the Holographic Principle and the AdS/CFT correspondence, which suggest that spacetime and gravity themselves may be emergent phenomena encoded in the quantum entanglement structure of a lower-dimensional boundary theory [8,9,10,11]. On cosmological scales, the prevailing ΛCDM model successfully describes an expanding universe whose spatial geometry appears nearly flat according to precision observations [12,13,14], though evidence for large-scale cosmic rotations presents intriguing challenges to standard assumptions [15,16]. Collectively, these diverse perspectives—from discrete geometry and higher dimensions to holographic emergence—underscore a deepening suspicion that the traditional dichotomy between "space" and "what fills space" may be a low-energy construct, hinting at a more profound unity.

A recent study demonstrates that tachyon-space system, via a mechanism of the "yolk-shell" coupling, represents the most efficient method of energy storage in the universe—enabling tachyons at super-Planck densities to be stably confined within a Planck-scale space. [17]. The symmetry analysis reveals a deep duality between the field and the geometry in this model. The vibration eigenmodes of the imaginary-mass scalar field are, by construction, identical to the eigenmodes of the Laplace-Beltrami operator on the compact spatial manifold with a boundary. An internal perspective views these as excitations of a quantum field (the "matter" yolk). An external, geometrical perspective views them as vibrational modes of the shape and size of the confining space itself (the "shell"). In the yolk-shell picture, these are not two independent things but two complementary descriptions of the same, self-consistent physical entity: a flexible space whose geometry has dynamically adjusted to trap and stabilize an unstable field, with its oscillations manifesting simultaneously as field vibrations and geometric deformations. This leads to a foundational philosophical thesis guiding the present work: Space is flexible, plastic, and self-adaptively curved in response to internal matter-energy. There is no space devoid of matter (a negatively curved surface would directly collapse); nor is there matter independent of space (a naked singularity would lead to infinite density). The origin of space is neither a pre-existing empty stage nor an abstract, entropic mathematical illusion. Matter and space form a naturally, physically indivisible and inseparable whole. The macroscopic continuous spacetime we perceive is precisely an emergent condensate, formed through the topological phase transition, mutual fusion, and "freezing" of countless such "microscopic geometric envelopes" at low energies. In essence, space is the necessary "garment" that matter must "wear" in order to exist. Our universe appears large, flat, and stiff not because space is fundamentally rigid, but because all matter within it jointly maintains a stable, coherent geometric configuration on cosmological scales.

To tackle the highly nonlinear coupling between geometry and the tachyon field, we adopt a systematic, iterative framework. We begin by constructing a zeroth-order approximation: seeking a static, confined solution for the tachyon field within a fixed, flat (Minkowski) spacetime background, treating the Dirichlet boundary condition as an external geometric constraint that models confinement. The properties of this solution—particularly its zero integrated energy—motivate the core physical principles of our theory. We then, in a first-order approximation, feed this matter configuration back into Einstein's equations to study the linear gravitational response and the resulting spatial curvature, which embodies the preliminary 'flexing' of spacetime. This iterative approach, from decoupled dynamics to coupled feedback, allows us to progressively build the physical picture of a self-consistent geometry-matter bound state, whose full solution would be the fixed point of this iterative procedure.

Building upon the "tachyon-space" coupling framework and the philosophical perspective above, this paper aims to formalize and extend these ideas into a comprehensive "geometry-matter bound state" theory. We begin by rigorously deriving the static, spherically symmetric tachyon-geometric bound state, explicitly proving its zero-energy condition and analyzing the spatial curvature profile (Section II). A comprehensive linear stability analysis is then performed, demonstrating how the flexible geometry provides restorative forces that quench the tachyonic instability, yielding stable oscillatory modes (Section III). The core principles of stability and zero-energy balance are subsequently elevated to a first principle, leading to the formulation of a general "Geometry-Matter Bound State Theory" via a constrained variational action (Section IV). This framework is then naturally extended to open systems through a proposed "Extremum-Adaptation Principle," offering a potential unified field theory framework capable of incorporating Standard Model fields. Finally, we discuss the broader implications, potential experimental signatures, and future directions of this "flexible spacetime" paradigm.

II: Tachyonic Geometric Constraints, Zero-Energy Condition, and Dynamics in (*d*+*1*)-Dimensional Spacetime

This chapter constructs the physical picture and methodological framework for the theory of "flexible spacetime-matter bound states." The starting point of this work is the deep-seated unity of "geometry-matter": there is no absolute space independent of matter, nor matter independent of space. The instability of a field with a "virtual mass" (tachyonic) potential $V(\phi) = -\frac{1}{2}\mu^2\phi^2$ does not imply its divergence in a presupposed infinite volume. Instead, it indicates that the spacetime structure itself must dynamically shape itself, forming a self-confined, finite-extent "energy-domain bubble" that completely wraps the field inside. Because the exterior admits no classical spacetime manifold (hence no Cauchy data at infinity), the tachyon field cannot realize its usual linearized runaway as outward propagation to infinity. Any attempt to lower its negative potential must be accommodated within the compact domain by redistributing stress onto the only available degrees of freedom—the localized boundary layer / shell variables ($R,\sigma,\ldots$)and the interior field gradients. Therefore, the solution for the field no longer depends on asymptotic boundary conditions at infinity but must be determined by the boundary structure of the bubble itself. It should be noted that the "flexible spacetime" and the self-confined states described in this theoretical framework constitute a hypothetical construct, and are not a direct depiction of the known properties of our observable universe.

The "flexibility" of this spacetime manifests in three attributes: rigid shaping driven by the principle of minimum energy (tending towards the most stable spherical symmetry), compliant stabilization through deformation dissipation in response to perturbations, and the geometric tension that is the root of this process. When the interior of the "bubble" (containing the ϕ field) and the exterior (another phase or vacuum) reach the limit of adaptation and matching at a characteristic scale R, the field naturally decays at the boundary, manifesting as $\phi(R)\to 0$. This R is not an externally imposed parameter but an equilibrium scale output by the system through a process of mutual adaptation and energy extremization.

The concrete dynamics of the "flexible spacetime wrapping and stabilizing the tachyonic field" is precisely the reciprocal conversion and damped balance between the energy of the interior field and the deformation/tension energy of the external boundary layer, achieved through coupled oscillations. This oscillation ultimately converges to a minimum of the system's effective potential $V_{eff}(R,\phi_0)$. Therefore, Rand the field amplitude ϕ_0 are not arbitrary external parameters but are conjugate variables of this coupled system, whose equilibrium values are given by the extremum condition of V_{eff}. The aforementioned nodalization of the field at the boundary, $\phi(R)\rightarrow 0$, is precisely the geometric manifestation of the equilibrium configuration under the limit of adaptation, a necessary requirement for the self-enclosure of the "energy-domain bubble," and serves as the dynamical source term for subsequently introducing gravitational feedback (the "flexible" response).

To substantiate the above physical picture, this chapter will provide a complete mathematical derivation for its core argument: that geometric structures spontaneously lock the tachyonic (imaginary-mass) field and ensure its stability (details in Appendix A). We will systematically prove, in the context of a $(d+1)$-dimensional spacetime, that: 1) Dirichlet boundary conditions lead to the eigenstate quantization of the static tachyonic field and its radius; 2) The total energy of the system is strictly zero; 3) The spatial curvature distribution exhibits a specific configuration; 4) The lowest-energy excitation corresponds to stable geometric deformations (ellipsoidal vibrations); 5) The observational signatures of such vibrations.

To rigorously construct this "self-shaped wrapping" state, we adopt a bottom-up, iterative, and step-by-step approximation paradigm. This chapter focuses on the first step of this paradigm: the zeroth-order approximation. We strategically concentrate on the self-trapping tendency of the matter field, temporarily fixing the interior geometry of the "bubble" to the simplest spherically symmetric background ($ds^2=-dt^2+dr^2+r^2d\Omega^2$), in order to reveal the inherent tendency of the matter field to spontaneously form static, discrete, zero-energy bound configurations within a finite region. The "zero-energy condition" and "nodalization" characteristics revealed by this zeroth-order solution are the cornerstone for elevating to the universal framework of the "extremum-adaptation principle."

Within this framework, the system possesses complete intrinsic dynamics. The "negative pressure" (downward potential) of the interior tachyonic field continuously "pushes out" against the wrapping shell, while the shell's tension generates an inward restoring force. The two reach dynamic balance through the adjustment of the radius R, forcing a sustained coupled oscillation between the field and the shell. Since energy cannot escape this self-confined energy domain, this oscillation is intrinsic, an adaptive cycle with no energy leakage or dissipation. As the self-shaping process approaches an extremum, the system settles into a stable equilibrium configuration—its spatial form is a spherical nodal standing wave, and its mechanical manifestation is the static force balance at the boundary against the field's negative pressure. The destabilization of this equilibrium, the "rupture" or "fission" of the configuration, is precisely the natural dynamical process of the system transitioning to a new equilibrium state when parameters exceed critical values within this theoretical framework, not a defect of the theory.

Therefore, the zeroth-order approximation not only provides the starting point for calculation but also depicts a complete dynamical picture ranging from self-organizing wrapping and adaptive oscillation to possible phase transition leaps, laying a solid physical and mathematical foundation for subsequently introducing geometric feedback and realizing the complete theory of "flexible spacetime-matter bound states."

II.1 Static Solution, Geometric Constraint, and Stability Criterion

Considering ϕ for the tachyonic field, $V(\phi)= -(1/2)\mu^2\phi^2$ for its potential, $g_{\mu\nu}$ for the metric, and R for the scalar curvature of the d-dimensional space， a d-dimensional space (topology S^d) with the spatial part of the metric:

$$ds^2 = dr^2 + r^2 d\Omega_{d-1}^2$$

where $d\Omega_{d-1}^2$ is the standard metric on the *(d-1)*-sphere. We seek a static, spherically symmetric tachyonic field solution: $\phi=\phi(r)$.

In curved space, the static field satisfies the Klein-Gordon equation:

$$\nabla^2\phi = \frac{\partial V}{\partial \phi} = -\mu^2\phi$$

where ∇^2 is the Laplace operator in *d*-dimensional space. For spherical symmetry, it reads:

$$\nabla^2\phi = \frac{1}{r^{d-1}}\frac{d}{dr}\left(r^{d-1}\frac{d\phi}{dr}\right)$$

Substituting the potential derivative yields a modified Bessel equation:

$$\frac{1}{r^{d-1}}\frac{d}{dr}\left(r^{d-1}\frac{d\phi}{dr}\right) + \mu^2\phi = 0$$

Its general solution is:

$$\phi(r) = r^{-(d-2)/2}\left[AJ_{(d-2)/2}(\mu r) + BY_{(d-2)/2}(\mu r)\right]$$

where J_v and Y_v are Bessel functions of the first and second kind, respectively, with $v=(d-2)/2$.

To obtain physically meaningful finite solutions, we impose the following boundary conditions:

Internal Regularity: The field must be finite at the center $r\to 0$. Since $Y_v(x)$ diverges as $x\to 0$, we set $B=0$.

Dirichlet Boundary Condition (Geometric Constraint): At a finite radius $r=R$ (representing the "bubble wall" or decoherence interface), the field is forced to zero: $\phi(R)=0$. This condition mimics the encapsulation and confinement of the unstable field (the "yolk") by the flexible geometric structure (the "shell").

Applying $\phi(R)=0$, we obtain the eigenvalue equation:

$$J_{(d-2)/2}(\mu R)=0.$$

Let $j_{v,n}$ be the *n-th* positive zero of J_v. The static solution exists only for discrete values of the geometric radius R:

$$R_n = \frac{j_{(d-2)/2,n}}{\mu} \qquad 1$$

Equation (1) is the core stability criterion. The intrinsic instability of the tachyonic field is precisely compensated by the finite spatial size R. A stable static standing wave forms only when the geometric radius Rand the mass parameter μ satisfy this quantization condition. This rigorously proves that **the flexible geometric structure of space is a necessary condition for stabilizing the tachyonic field**. The ground-state ($n=1$) field configuration is:

$$\phi(r) = \phi_0 \frac{J_v(\mu r)}{J_v(\mu R)} \quad (0\le r\le R).$$

where ϕ_0 is the central field value.

II.2 First-Principles Proof of the Zero-Energy Condition

To verify the total energy is zero, we calculate the integrated energy of the static solution within a *d*-dimensional ball of radius R. We work in a *(d+1)*-dimensional Minkowski background metric $ds^2 = -dt^2 + dr^2 + r^2 d\Omega_{d-1}^2$. The energy-momentum tensor for the tachyonic field is:

$$T_v^\mu = \partial^\mu\phi\partial_v\phi - \delta_v^\mu\left[\frac{1}{2}(\nabla\phi)^2 + V(\phi)\right]$$

For the static, spherically symmetric solution, the energy density is:

$$\rho(r) = T_0^0 = \frac{1}{2}\left(\frac{\mathrm{d}\phi}{dr}\right)^2 - \frac{1}{2}\mu^2\phi^2$$

The total energy is obtained by integrating over the entire space:

$$E_{total} = \int_0^R \rho(r)dV = \int_0^R \left[\frac{1}{2}\left(\frac{d\phi}{dr}\right)^2 - \frac{1}{2}\mu^2\phi^2\right]dV$$

where $dV=r^{d-1}drd\Omega_{d-1}$ is the volume element.

Using integration by parts and the field's equation of motion, the kinetic term can be rewritten:

$$\int_0^R \frac{1}{2}\left(\frac{\mathrm{d}\phi}{dr}\right)^2 dV = -\frac{1}{2}\int_0^R \phi(\nabla^2\phi)dV + \frac{1}{2}\oint_{S^{d-1}} \phi\frac{d\phi}{dr}\mathrm{dS} = \frac{1}{2}\int_0^R \mu^2\phi^2 dV + \frac{1}{2}\oint_{S^{d-1}} \phi\frac{d\phi}{dr}\mathrm{dS}$$

The field equation $\nabla^2\phi=-\mu^2\phi$ was used. The boundary term is an integral over the *(d-1)*-sphere S^{d-1}. Imposing the Dirichlet boundary condition $\phi(R)=0$ and considering regularity at the origin $r=0$, this boundary term vanishes exactly. Therefore,

$$E_{total} = \frac{1}{2}\int_0^R \mu^2\phi^2 dV - \frac{1}{2}\int_0^R \mu^2\phi^2 dV = 0$$

This proof is independent of the specific dimension *d*. It rigorously establishes that, under Dirichlet boundary conditions, the negative potential energy of the tachyonic field and its gradient kinetic energy (corresponding to geometric confinement energy) cancel exactly, resulting in a system with zero total energy. This provides a first-principles mathematical foundation for the main text's assertion of a "locked balance between matter (the "yolk") and geometry (the "shell")."

II.3 Distribution of the Spatial Curvature Scalar

Using the zeroth-order field solution $\phi(r)$ from Sec. II.1 as a source, we now examine the linear gravitational response of the spacetime geometry via the Einstein field equations, which constitutes the first-order adaptation of the geometry to the matter distributionTo investigate how space curves in response to the tachyonic field, we examine the interior geometry via the Einstein equations. In (*d+1*)-dimensional spacetime, the Einstein equations are:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \kappa_{d+1}T_{\mu\nu}$$

Taking the trace gives the relation between the Ricci curvature scalar Rand the trace of the energy-momentum tensor $T=T_\mu^\mu$:

$$\mathrm{R} = -\frac{2}{d-1}\kappa_{d+1}T$$

For the tachyonic field, the trace is:

$$T = -(d-1)\cdot\frac{1}{2}(\nabla\phi)^2 + d\cdot V(\phi) = -(d-1)\cdot\frac{1}{2}(\nabla\phi)^2 - \frac{d}{2}\mu^2\phi^2$$

For a concrete analysis, consider a three-dimensional space (d=3). Here, the static solution is $\phi(r)=A\sin(\mu r)/r$. Substituting yields:

$$T(r) = -\left(\frac{d\phi}{dr}\right)^2 - \frac{3}{2}\mu^2\phi^2$$

The curvature scalar $R(r) \propto -T(r)$. Analyzing its distribution:

Near the center (r→0): $\phi \approx constant$, $\nabla\phi \approx 0$, so $T<0$ and thus $R>0$, indicating positive curvature.

Near the boundary (*r→R*): $\phi(R)=0$, but the gradient $\nabla\phi$ is large, making T negative with a large magnitude, and R positive and significant. This indicates stronger spatial curvature in the boundary region due to the rapid variation of the field.

This calculation reveals how the flexible space curves in response to and encapsulates the tachyonic field: the interior region has positive curvature, with a more pronounced curvature variation at the boundary. The geometric configuration and field distribution are mutually locked, jointly maintaining the system's zero-energy state.

II 4 Dynamics of Ellipsoidal Deformation

Breaking perfect spherical symmetry, we consider the lowest-energy excitation—the quadrupole deformation (ellipsoidal shape). Expand the field as:

$$\emptyset(t, r, \theta) = \emptyset_0(r) + \varepsilon T(t) \cdot Y_{l=2}(\theta)$$

where $Y_{l=2}$ is the d-dimensional spherical harmonic for the quadrupole mode (*l=2*), and ϵ is a small parameter. Substituting into the time-dependent Klein-Gordon equation and separating variables yields the equation for the shape vibration mode *T(t)*:

$$\ddot{T} + \left[\frac{l(l + d - 2)}{R^2} - \mu^2\right] T = 0$$

Defining the vibration frequency ω:

$$\omega^2 = \frac{l(l+d-2)}{R^2} - \mu^2 \qquad 2$$

Using the stability criterion for the static solution, $\mu R = j_{(d-2)/2,n}$, and noting that for $l \geq 2$, typically $l(l + d - 2) > j^2_{(d-2)/2,1}$, we find $\omega^2 > 0$. This means the lowest-energy deformation mode is a stable harmonic oscillation $T(t) \sim cos(\omega t)$, not a collapse**.** This corresponds to the "flexible geometric deformation (membrane vibration)" in the main text. During vibration, the kinetic energy of the geometry and the potential energy of the tachyonic field convert into each other, with the zero-energy condition ensuring $E_{total} \equiv 0$. Equation (A.2) directly links the geometric characteristic scale *R*, the field's mass parameter μ, and the vibration frequency ω, again demonstrating the fundamental constraint geometry imposes on the system's dynamics.

II.5 Time Evolution and Connection to Experimental Observation

Incorporating the time factor $e^{i\omega t}$, the dynamic curvature scalar also varies periodically:

$$R(t, r, \theta) = R_0(r) + \delta R cos(\omega t + \varphi)$$

This constitutes the "ripple" of the higher-dimensional geometry. For an external observer in (3+1)-dimensional spacetime, this vibration projects as an effective time-varying Newtonian potential perturbation $\delta U(t)$. In experiments like atom interferometry, this perturbation modulates the matter-wave phase:

$$\delta\phi(t) \approx \frac{1}{\hbar} \int m\delta U(t)\, dt \propto Y_{l,m}(\theta, \varphi) cos(\omega t + \varphi_0)$$

The vibration frequency ω, determined by the geometric scale *R* and mass parameter μ, can reach the order of 10^{43}Hz at the Planck scale. Practical instruments average over this high-frequency signal, registering it as phase noise within a specific bandwidth. This signal may possess anisotropic features (dictated by the higher-dimensional spherical harmonics $Y_{l,m}$) and non-Gaussian statistical characteristics, providing potential fingerprints for experimental identification.

III: Linear Stability Analysis: Feedback and Stabilization via Flexible Geometry

This chapter aims to perform a linear perturbation analysis on the static, spherically symmetric, zero-total-energy "Tachyon-Geometric Bound State" solution proposed in the main text. The core objective is to test the behavior of this equilibrium solution under small dynamical perturbations, determining whether it diverges (unstable) or oscillates (stable).

III.1 Background Solution and Perturbation Setup

Background Spacetime: Employ a $d+1$ dimensional spacetime metric (1 time, d space dimensions), with the background being the static, spherically symmetric solution:

$$ds_0^2 = -f_0(r)dt^2 + h_0(r)dr^2 + r^2 d\Omega_{d-1}^2$$

where $f_0(r)$ and $h_0(r)$ are determined jointly by the Einstein field equations and the Tachyon field equation ($V(\phi)=-\frac{1}{2}\mu^2\phi^2$), corresponding to the static solution $\phi_0(r)$ derived in Appendix A, which satisfies the Dirichlet boundary condition $\phi(R)=0$, and its associated geometry.

Background Field: The Tachyon field background is this static solution: $\phi_0=\phi_0(r)$.

Introduction of Perturbations: Introduce arbitrary but small, time-dependent perturbations onto the background solution:

$$g_{\mu\nu}(t,r,\theta) = g_{\mu\nu}^0(r) + \epsilon h_{\mu\nu}(t,r,\theta)$$
$$\phi_{\mu\nu}(t,r,\theta) = \phi_{\mu\nu}^0(r) + \epsilon\delta\phi(t,r,\theta)$$

where $\epsilon \ll 1$ is a small parameter, $h_{\mu\nu}$ is the metric perturbation, and $\delta\phi$ is the field perturbation. This analysis retains terms up to first order in $O(\epsilon)$.

III.2 Total Action and Its Second-Order Variation

The total action of the system is the sum of the Einstein-Hilbert action and the matter field action:

$$S_{total} = S_{EH} + S_\phi = \int d^4x\sqrt{-g}\left(\frac{1}{2\kappa}R + L_\phi\right)$$

where $\kappa=8\pi G$ and $L_\phi = \frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\upsilon\phi - V(\phi)$. Linear stability is determined by the second-order variation of the total action about the background solution.

Second-Order Variation of the Gravitational Part: $\delta^{(2)}S_{EH}$ measures the "elastic potential energy" of spacetime geometry deviating from the background. Standard computation yields:

$$\delta^{(2)}S_{EH} = \frac{1}{2\kappa_D}\int d^4x\sqrt{-g_0}h_{\mu\nu}\varepsilon^{\mu\nu\rho\sigma}h_{\rho\sigma}$$

where $\varepsilon^{\mu\nu\rho\sigma}$ is a second-order differential tensor operator constructed from the background metric $g_{\mu\nu}^{(0)}$ and its derivatives.

Second-Order Variation of the Matter Part: $\delta^{(2)}S_\phi$ includes the kinetic and potential energies of the field perturbation itself, as well as the interaction energy between the perturbation and the background field:

$$\delta^{(2)}S_\phi = \int d^4x\sqrt{-g_0}\left[-\frac{1}{2}\nabla_\mu\delta\phi\nabla^\mu\delta\phi + \frac{1}{2}\mu^2(\delta\phi)^2\right] + coupling\ terms$$

The crucial coupling terms are embodied in the first-order variation of the energy-momentum tensor $\delta T_{\mu\nu}$, which depends linearly on $h_{\mu\nu}$ and $\delta\phi$. This is the mathematical manifestation of the **self-regulating feedback mechanism**: the field perturbation $\delta\phi$ alters the local energy-momentum distribution ($\delta T_{\mu\nu}$), thereby driving a response in the spacetime geometry ($h_{\mu\nu}$); conversely, the geometric deformation ($h_{\mu\nu}$) also reacts back on the field evolution through geodesic deviation effects.

Coupled Linear Perturbation Equations: Varying the total action $\delta^{(2)}S_{total}$ with respect to $h_{\mu\nu}$ and $\delta\phi$ separately yields a set of coupled linear equations:

Geometric Perturbation Eqn: $\varepsilon_{\mu\nu\rho\sigma}h^{\rho\sigma} = \kappa_D\delta T_{\mu\nu}(\phi_0,\delta\phi)$

Field Perturbation Eqn: $(\Box_0 - \mu^2)\delta\phi = S(h_{\alpha\beta};\phi_0)$

Here, $\Box_0$ is the d'Alembertian operator on the background metric, and S is the source term generated by the metric perturbation hαβin the field equation. The two equations must be solved simultaneously.

III.3 Separation of Variables and the Eigenvalue Problem

The static, spherical symmetry of the background allows for a spectral decomposition of the perturbations, transforming the stability problem into an eigenvalue problem.

1. **Harmonic Time Dependence and Angular Decomposition:** Given the static background, we can assume perturbations have a harmonic time dependence $e^{-i\omega t}$. Furthermore, exploiting spherical symmetry, perturbations are expanded in terms of d-dimensional spherical harmonics $Y_{\ell m}(\theta)$. To investigate the most unstable modes, one typically first analyzes the spherically symmetric "breathing mode" (i.e., $\ell=0$ scalar-type perturbations):

$$h_{\mu\nu}(t,r) = e^{-i\omega t}\tilde{h}_{\mu\nu}(r)$$
$$\delta\phi(t,r) = e^{-i\omega t}\tilde{\chi}(r)$$

2. **Reduction to a Radial Eigenvalue Problem:** Substituting the above forms into the coupled perturbation equations and using constraints (e.g., the Regge-Wheeler gauge) to eliminate non-dynamical degrees of freedom, the complex tensor equations can be reduced to one or a set of coupled second-order ordinary differential equations for the radial functions $\tilde{\chi}(r)$ and certain metric perturbation components (e.g., $\tilde{h}_{tt}(r)$, $\tilde{h}_{rr}(r)$). After suitable variable choices and coordinate transformations (such as introducing the tortoise coordinate $\mathrm{dr}_* = \sqrt{\mathrm{h}_0/\mathrm{f}_0}\mathrm{dr}$), the system can be written compactly as:

$$\left(\frac{d^2}{dr_*^2} + \omega^2\right)\begin{pmatrix}\tilde{\chi}\\ \tilde{h}\end{pmatrix} = V(r;\phi_0, f_0, h_0)\begin{pmatrix}\tilde{\chi}\\ \tilde{h}\end{pmatrix}$$

Here, $V(r)$ is a 2×2 potential matrix, all of whose matrix elements are determined by the static background solution $\phi_0(r)$, $f_0(r)$, $h_0(r)$ and their derivatives. This equation is analogous in form to a coupled-channel Schrödinger equation in quantum mechanics, with ω^2 as the eigenvalue.

III.4 Stability Criterion and Physical Conclusion

The dynamical stability of the system is entirely determined by the nature of the eigenvalue problem for ω^2:

If $\omega^2>0$ for all perturbation modes, then ω is real, and perturbations represent stable oscillations about the equilibrium position (i.e., eigenmodes are $e^{-i\omega t}$ with real ω).

If there exists any mode with $\omega^2<0$, then ω is imaginary, and the perturbation will grow exponentially in time ($e^{|\omega|t}$), indicating linear instability of the system.

In the present theoretical context, the following key stabilization mechanisms are at play:

Geometric Confinement and Dirichlet Boundary Condition: The background solution $\phi_0(r)$ derived in chapter II is itself an eigenstate of the Tachyon field within a finite geometry (radius R). The Dirichlet boundary condition $\phi(R)=0$ provides a powerful external constraint, suppressing unlimited growth of the field.

Zero-Energy Condition and Self-Regulating Feedback: The total energy of the system is strictly zero ($E_{total}=0$). This implies that any field perturbation $\delta\phi$ attempting to increase (or decrease) its negative potential energy will necessarily and instantly excite a geometric response $h_{\mu\nu}$ (i.e., deformation of spacetime) of appropriate magnitude and opposite sign, providing a restoring force via the gravitational interaction $\delta T_{\mu\nu}$ to conserve total energy. Mathematically, this manifests as strong off-diagonal coupling terms in the potential matrix $V(r)$.

Therefore, it is physically expected that the intrinsic instability of the Tachyon field ($\mu^2<0$) is overcome by the restorative force of the flexible space. Mathematically, this predicts that under the influence of the potential matrix $V(r)$, the eigenvalues ω^2 will be shifted upward from their original negative value ($-|\mu^2|$) to become positive real numbers. Solving the eigenvalue problem (typically requiring numerical methods) would yield the specific spectrum of stable oscillation frequencies.

Based on the core physical picture of "spatial flexibility" and the "zero-energy constraint," the linear perturbation theory framework constructed here indicates that the static "Tachyon-Geometric Bound State" is not merely an equilibrium solution to the Einstein and Tachyon field equations, but is also expected, upon introducing perturbations, to transform unstable exponential growth modes into stable geometric vibrational modes via the self-coupling feedback mechanism between geometry and matter. However, this theoretical framework is mathematically self-consistent and strongly suggests dynamical stability.

IV. The Zero-Energy Principle and a Unified Geometry-Matter Framework

IV.1 The Zero-Energy Constraint as a First Principle

The stable, zero-energy tachyon-geometric bound state derived in Chapters II and III is not merely a special solution but exemplifies a profound underlying principle. We now elevate this core insight to the status of a **fundamental postulate**: in a self-consistent, finite physical system, the total energy of the coupled geometry-matter entity must vanish. This is the **Zero-Energy Principle**.
Mathematically, for a universe treated as a closed quantum-gravitational system, the Wheeler-DeWitt equation $\hat{H}|\psi\rangle = 0$ enforces this at the quantum level. In a semi-classical continuum description, the principle mandates that the expectation value of the total Hamiltonian vanishes:

$$\langle \hat{H}_{total} \rangle = \langle \hat{H}_g \rangle + \langle \hat{H}_m \rangle = 0 \qquad 3$$

where $\langle \hat{\boldsymbol{H}}_{\boldsymbol{m}} \rangle$ and $\langle \hat{\boldsymbol{H}}_{\boldsymbol{g}} \rangle$ are the expectation values of the matter and geometric (gravitational) energy, respectively. This condition is not an ad-hoc addition but arises from the diffeomorphism invariance of a closed universe. It encapsulates the perfect, locked balance between matter (whose excitations can carry negative energy density, as in the tachyonic potential) and geometry (whose deformation carries positive energy). The Zero-Energy Principle thus serves as the foundational axiom from which a unified theory of geometry and matter can be constructed.

IV.2 Geometry-Matter Bound State Theory for Closed Systems

To formalize the theory for closed, self-contained systems where the zero-energy condition (3) holds strictly, we construct a dynamical framework based on a **constrained variational principle** (Details in Appendix B). The core object is the **meta-action** S, which incorporates the fundamental dynamical variables—the metric field $g^{\mu\nu}$ and the matter field ϕ—along with a Lagrange multiplier term that enforces the global geometry-matter duality:

$$S = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} + L_m(\phi)\right] + \lambda \cdot C[\phi,\ g] \qquad 4$$

Here, $C[\phi,g]=0$ is the **duality constraint equation**, an algebraic or functional relation that physically represents the topological compromise between the radial (geometric) and angular (matter) degrees of freedom, enforcing the zero-energy condition. The parameter λ is a Lagrange multiplier; in the strong-coupling limit relevant for a perfectly closed system, $\lambda\to\infty$, making $C=0$ a rigid constraint. The canonical matter Lagrangian is, for example for a tachyonic field, $L_m = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi + \frac{1}{2}\mu^2\phi^2$.

The dynamics are derived from the stationary action principle *δS=0*. Variation yields a set of four tightly coupled equations that define the theory:

The Duality Constraint: This is the defining axiom of the theory, recovered from varying with respect to λ:

$$C[\phi, g] = 0 \qquad \mathbf{5}$$

This general framework is self-consistent with the specific tachyonic field bound-state model constructed earlier in Section II. In fact, that concrete model can be viewed as a particular solution of this framework in the strong-coupling limit ($\lambda \to \infty$). For this model, a natural and concrete realization of the adaptation functional $C[\phi,g]$ is the total energy of the system:

$$C[\phi, g] = E_{total}[\phi, g] = \int_V dV \left[\frac{1}{2}(\nabla\phi)^2 - \frac{1}{2}\mu^2\phi^2\right].$$

The constraint $C[\phi,g]=0$ then forces the system into a state of zero total energy. As we proved in Section II.2, under the Dirichlet boundary condition $\phi|_{\partial V}=0$, this condition is automatically satisfied. This boundary condition itself, combined with the field equation, leads to the eigenvalue condition $J_{(d-2)/2}(\mu R)=0$ that determines the stable configuration of the system, or its approximate form for specific excitation modes, $\mu^2 \approx l(l+d-2)/R^2$. Therefore, in the tachyonic field model, the physical content of the constraint equation $C[\phi,g]=0$ is concretely manifested as zero total energy, and its mathematical consequence is the stability condition characterizing the locked relationship between the geometric scale R and the field parameter μ. This verifies our framework: taking $C[\phi,g]=0$ as a first principle can lead to a self-consistent, stable geometry-matter bound state. This provides confidence and a concrete paradigm for us to extend this principle to more general matter fields (see **IV.3**).

The Meta-Action: The total action (4) itself unifies the description of geometry, matter, and their coupling.

The Modified Einstein Equation (Geometric Response): Varying with respect to the metric $g^{\mu\nu}$ gives:

$$G_{\mu\nu} = 8\pi G\left(T_{\mu\nu}^{(m)} + \lambda\frac{\delta C}{\delta g^{\mu\nu}}\right) \qquad \mathbf{6}$$

This shows that spacetime curvature is sourced not only by the canonical matter stress-energy tensor $\boldsymbol{T_{\mu\nu}^{(m)}}$ but also by a geometric shaping force arising from the variation of the constraint. This force allows space to flexibly adapt its curvature to match the matter content.

The Unified Matter Equation: Varying with respect to the matter field ϕ yields:

$$(\Box - \boldsymbol{\mu^2})\boldsymbol{\phi} + \frac{\boldsymbol{\partial L_{int}}}{\boldsymbol{\partial\emptyset}} = -\boldsymbol{\lambda}\frac{\boldsymbol{\delta C}}{\boldsymbol{\delta\phi}} \qquad \mathbf{7}$$

This single equation governs both tachyonic ($\mu^2<0$) and massive ($\mu^2>0$) fields. The stability of either is ensured by the balancing force on the right-hand side, which originates from the constraint.

The quadruplet of equations (7) forms a closed, self-consistent system defining the Geometry-Matter Bound State Theory for closed systems. In the strong-coupling limit ($\lambda\to\infty$), the constraint $C=0$ dominates, forcing the system into configurations where geometry and matter are locked. The static, spherically symmetric tachyon-geometric solution of Chapter II emerges as the ground-state solution of this constrained theory.

IV.3 The Extremum-Adaptation Principle: A Unified Field Theory Framework for Open Systems

IV.3.1 The Extremum-Adaptation Principle and the Total Action

The rigid zero-energy constraint of Section 4.2 describes an ideal, isolated universe. To describe subsystems, excited states, or systems interacting with an environment—where the total energy is not strictly zero and the geometry-matter coupling is finite—we must generalize the framework. We propose the **Extremum-Adaptation Principle** as the first principle for autonomous spacetimematter systems: the physical configuration $(g_{\mu\nu}, \Phi)$ is determined by the stationary action principle $\delta S_{total}[g, \Phi]=0$ of a total action that includes a measure of the system's tendency to achieve an optimal, stable configuration.

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} + L_m(\phi, g) \right] + \beta \cdot C[\phi, g] \tag{8}$$

The terms are interpreted as follows:

R/(16πG): The Einstein-Hilbert term, describing the pure spacetime geometry dynamics.

Lm(Φ,g): The matter field Lagrangian density, describing the dynamics of matter fields Φ in the geometric background g. Here, Φ is a generalized variable representing all matter fields.

C[Φ,g]: The **adaptation functional**. This is the core innovation of the theory, quantitatively encoding the system's propensity for "spacetime geometry and matter fields to mutually adapt towards a globally optimal, stable state." Its specific physical form (e.g., related to total system energy, information complexity, or stability measure) is a primary subject for future research. β: A coupling constant with appropriate dimensions. It characterizes the strength of the "adaptation tendency" and is a new fundamental parameter of the theory.

IV.3.2 The General Extremum-Adaptation Equations (hypothesis)

Performing independent variations of the total action (8) with respect to the metric $g^{\mu\nu}$ and the matter field Φ yields the fundamental, coupled dynamical equations.

The Generalized Einstein Equation (*from* $\delta S_{total}/\delta g^{\mu\nu}=0$):

$$G_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^{(m)} + \beta \frac{\delta C}{\delta g^{\mu\nu}} \right) \tag{9}$$

Here, $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}$ is the Einstein tensor. $T_{\mu\nu}^{(m)} = -\frac{2}{\sqrt{-g}} \frac{\delta \sqrt{-g} L_m}{\delta g^{\mu\nu}}$ is the classical stress-energy tensor of the matter fields. The new term, $\beta \cdot \delta C/\delta g^{\mu\nu}$, is the adaptation shaping tensor. It represents an active shaping force exerted on spacetime geometry by the matter fields through the adaptation tendency C, allowing geometry to flexibly respond.

The Generalized Matter Field Equation (from $\delta S_{total}/\delta \Phi=0$):

$$[\text{Standard Matter Field Equation}] = -\beta \cdot \frac{\delta C}{\delta \Phi} \tag{10}$$

The left-hand side [⋯]corresponds to the standard Euler-Lagrange equation for the field Φ in curved spacetime (e.g., Klein-Gordon, Yang-Mills, Dirac equation). The right-hand side, $-\beta \cdot \delta C/\delta \Phi$, is the new **geometric feedback term**. It represents the constraint and feedback exerted on the matter field's evolution by the spacetime geometry, mediated by the adaptation tendency C.

The coupled system of equations (9) and (10), linked through the adaptation functional C and its variational derivatives, must be solved self-consistently. They jointly define the "spacetime-matter adaptation state."

IV.3 Application to Fundamental Fields of the Standard Model

Applying this general framework to the Lagrangian of the Standard Model yields a unified set of modified field equations for all known interactions, demonstrating the framework's universality.

Scalar Field (with mass m^2, which can be positive for a standard field or negative $m^2=-\mu^2$ for a tachyon): For a scalar field ϕ with $L_m = -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - \frac{1}{2}m^2\phi^2$, the extremum-adaptation equation is the modified Klein-Gordon equation:

$$(\square - m^2) = -\beta \cdot \frac{\delta C}{\delta \phi} \tag{11}$$

Electromagnetic Field (*U(1)* gauge field): For the gauge potential A_μ with $L_m = -\frac{1}{4}\mathrm{F}_{\mu\nu}\mathrm{F}^{\mu\nu}$, we obtain the modified Maxwell equations:

$$\nabla_\mu \mathrm{F}^{\mu\nu} = -\beta \cdot \frac{\delta \mathrm{C}}{\delta \mathrm{A}_\nu} \tag{12}$$

Electroweak Field ($SU(2)_L \times U(1)_Y$ gauge fields + Higgs field):

For the fields $\Phi = \{\mathrm{W}_\mu^i, \mathrm{B}_\mu, \mathrm{H}\}$, where H is the Higgs doublet, the extremum-adaptation equations are:

$$\left(\mathrm{D}_\mu \mathrm{W}^{\mu\nu}\right)^i = -\beta \cdot \frac{\delta \mathrm{C}}{\delta \mathrm{W}_\nu^i} \quad \text{.(Modified } SU(2)_L \text{ Yang-Mills)}$$

$$\partial_\mu B^{\mu\nu} = -\beta \cdot \frac{\delta \mathrm{C}}{\delta \mathrm{B}_\nu} \quad \text{(Modified U(1)Y Maxwell)} \tag{13}$$

$$\left(D_\mu D^\mu H\right) - \frac{\partial V}{\partial H^\dagger} = -\beta \cdot \frac{\delta C}{\delta H^\dagger} \quad \text{(Modified Higgs Equation)}$$

Within this framework, electroweak symmetry breaking and mass generation can be interpreted as the system's spontaneous process of finding a stable vacuum solution, driven by the adaptation tendency C.

Strong Interaction Field ($SU(3)_c$ gauge field):

For the gluon field G_μ^a , the theory yields the modified Quantum Chromodynamics (QCD) Yang-Mills equation:

$$\left(D_\mu G^{\mu\nu}\right)^a = -\beta \cdot \frac{\delta C}{\delta G_\nu^a} \tag{14}$$

The new term on the right-hand side may provide a novel geometric perspective on the microscopic mechanisms of color confinement and asymptotic freedom.

Therefore, the "Zero-Energy Constraint Framework" describes the very special state of the entire universe as an isolated, closed system in a state of absolute energy balance. In contrast, the present "Extremum-Adaptation Framework" universally describes open systems—from microscopic particles to macroscopic astrophysical objects—as they evolve, under the combined influence of their intrinsic dynamics and internal adaptation tendency, towards some optimal stable configuration. This establishes a clear hierarchy where the specific, rigidly constrained models are natural limits of the more general, flexible theory. It should be stated in the main text that while a simple example construction of *C[Φ,g]* is presented in Appendix C6 (following the principles outlined in Appendix C5), the complete explicit construction for arbitrary cases remains an open challenge.

V Summary and Outlook

This paper establishes a complete theoretical system for "flexible spacetime," progressing from concrete models to universal principles. The core paradigm is **geometry-matter duality**: spacetime is not a static background but a dynamic entity that can form self-consistent bound states with matter fields. This duality is microscopically realized in the specific "tachyon-geometric bound state" solution, where the finite size and flexible curvature of space precisely cancel the tachyonic instability via the **Zero-Energy Principle**—a perfect locked balance between matter (negative energy) and geometry (positive energy), resulting in a system with strictly vanishing total energy. Macroscopically and generally, this balancing mechanism is elevated to the **Extremum-Adaptation Principle**. The co-evolution of spacetime and matter is governed by the extremization of an action containing an "adaptation functional" $C[g,\Phi]$. This principle naturally yields unified, modified Einstein and matter field equations, placing gravity and Standard Model interactions within a single geometric framework. The deep philosophical foundation lies in the **spacetime-matter topological duality**: the continuous longitudinal (scale) degrees of freedom and the discrete angular (topological) degrees of freedom constrain each other. This explains why the universe appears as a rigid, flat space at low energies, while exhibiting intense geometric flexibility at high energies (e.g., the Planck scale). This theory provides a novel framework for understanding quantum gravity, the origin of the universe (e.g., replacing the initial singularity), and the problem of fundamental constants. Future work will focus on (i) converging the iterative procedure by solving the fully coupled Φ–g system to obtain the exact bound-state solution, and (ii) upgrading the adaptation functional $C[\Phi,g]$—whose construction principles are outlined in Appendix C5 with an illustrative example in Appendix C6—into a physically grounded operator. This includes identifying its most plausible microscopic origin (e.g. quantum-information–theoretic constraints or topological invariants), fixing ambiguities/regularization, and deriving model-dependent predictions such as ultra-high-frequency “spacetime ripples.” The goal is then to confront those predictions with precision laboratory measurements and cosmological observations to enable an empirical test of the Extremum-Adaptation Framework.

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Appendix A: Detailed Derivation of Linear Perturbation Analysis

This appendix provides the complete mathematical derivation for the linear stability analysis of the static, spherically symmetric tachyon-geometric bound state. The goal is to derive the coupled eigenvalue problem that determines the stability of the solution presented in Chapter II.

A.1 Background Solution and Perturbation Setup

We consider the system of a tachyonic scalar field ϕ with potential $V(\phi) = -\frac{1}{2}\mu^2\phi^2$ minimally coupled to gravity in a *d+1*dimensional spacetime. The action is

$$S = \int d^{d+1}x\sqrt{-g}\left[\frac{1}{2\kappa_d}R + \frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi)\right]$$

where $\kappa_d = 8\pi G_{d+1}$.

The background is the static, spherically symmetric bound state solution $\bar{g}_{\mu\nu}(r), \bar{\phi}(r)$ derived in in the main text. The background metric is

$$ds^2 = -\bar{A}(r)dt^2 + \bar{B}(r)dr^2 + r^2 d\Omega_{d-1}^2 \qquad A1$$

and the background field $\bar{\phi}(r)$ satisfies the boundary condition $\bar{\phi}(R) = 0$. The overbar denotes a background quantity.

We introduce linear, time-dependent perturbations to this background:

$$g_{\mu\nu}(t,x) \; = \bar{g}_{\mu\nu}(r) + \epsilon h_{\mu\nu}(t,x)$$
$$\phi(t,x), = \bar{\phi}(r) + \epsilon\delta\phi(t,x) \qquad A2$$

where $\epsilon \ll 1$. The metric perturbation $h_{\mu\nu}$ and field perturbation $\delta\phi$ are the dynamical variables whose evolution determines stability.

A.2 Linearized Field Equations

The linearized dynamics are governed by the equations obtained from the second-order variation of the action, $\delta^{(2)}S=0$.

1. Linearized Einstein Equations:

Varying with respect to the metric gives

$$\delta G_{\mu\nu} = \kappa\delta T_{\mu\nu} \qquad A3$$

where $\delta G_{\mu\nu}$ is the linear perturbation of the Einstein tensor, and $\delta T_{\mu\nu}$ is the linear perturbation of the scalar field stress-energy tensor, Its explicit form couples $h_{\mu\nu}$ and $\delta\phi$:

$$\delta T_{\mu\nu} = \partial_\mu\bar{\phi}\partial_\nu(\delta\phi) + \partial_\nu\bar{\phi}\partial_\mu(\delta\phi) -$$
$$\bar{g}_{\mu\nu}\left[\bar{g}^{\alpha\beta}\partial_\alpha\bar{\phi}\partial_\beta(\delta\phi) + \mu^2\bar{\phi}\delta\phi\right] - h_{\mu\nu}\left[\frac{1}{2}(\bar{\nabla}\bar{\phi})^2 + V(\bar{\phi})\right] + \cdots \qquad A4$$

2. Linearized Klein-Gordon Equation:

Varying with respect to the scalar field yields

$$\bar{\Box}(\delta\phi) - \mu^2\delta\phi = S\left(h_{\alpha\beta};\bar{\phi}\right) \qquad A5$$

where $\bar{\Box}$ is the d'Alembertian operator on the background. The source term S is generated by the metric perturbation:

$$S(h_{\alpha\beta};\bar{\phi}) = -\frac{1}{\sqrt{-\bar{g}}}\partial_\mu\left(\frac{1}{2}\sqrt{-\bar{g}}h\bar{g}^{\mu\nu}\partial_\nu\bar{\phi}\right) + \bar{g}^{\mu\nu}\partial_\nu\bar{\phi}\partial_\mu\left(\frac{\hbar}{2}\right) + \cdots$$

And $h = \bar{g}^{\mu\nu}h_{\mu\nu}$.

A.3 Mode Decomposition and Gauge Fixing

Given the spherical symmetry and static nature of the background, we decompose perturbations into spherical harmonics $Y_{lm}(\theta)$ and assume a harmonic time dependence $e^{-i\omega t}$. For the most general analysis, one must consider perturbations of different parities (polar and axial). The most relevant mode for stability against collapse or dispersion is the spherically symmetric **polar (even-parity) "breathing" mode** (*l=0*). To eliminate gauge degrees of freedom, we adopt the **Regge-Wheeler gauge** for polar perturbations. In this gauge, the metric perturbation $h_{\mu\nu}$ is expressed through a limited set of master functions. For a spherically symmetric background, the even-parity metric perturbations can be parameterized by two functions, $H_0(r)$ and $H_1(r)$, in the time-radius sector, and a function $K(r)$ related to the angular part. After imposing this gauge and substituting the harmonic decompositions $h_{\mu\nu}(t,r) = e^{-i\omega t}\tilde{h}_{\mu\nu}(r)$ and $\delta\phi(t,r) = e^{-i\omega t}\tilde{\chi}(r)$, the system of partial differential equations reduces to a set of coupled ordinary differential equations (ODEs) for the radial profiles.

A.4 Reduction to a Coupled Schrödinger-Type System

Through a series of manipulations—using the constraint equations from the *(θ,θ)* and *(t,r)* components of the linearized Einstein equations, and introducing suitable master variables—the system can be cast into a standard form. A common approach is to define a geometric master variable $\Psi_g(r)$ (a Zerilli-like function) and retain the field perturbation variable $\Psi_\phi(r) = \tilde{\chi}(r)$. After introducing the tortoise coordinate $dr_* = \sqrt{\bar{B}(r)/\bar{A}(r)}\,dr$, the final coupled radial equations take the compact matrix form:

$$\left(-\frac{d^2}{dr_*^2}\boldsymbol{I} + V(r)\right)\begin{pmatrix}\Psi_g(r)\\ \Psi_\phi(r)\end{pmatrix} = \omega^2\begin{pmatrix}\Psi_g(r)\\ \Psi_\phi(r)\end{pmatrix} \qquad \text{A6}$$

where I is the 2×2 identity matrix. This is a coupled-channel eigenvalue problem for ω^2. The 2×2 potential matrix V(r) is

$$V(r) = \begin{pmatrix}V_{gg}(r) & V_{g\phi}(r)\\ V_{\phi g}(r) & V_{\phi\phi}(r)\end{pmatrix} \qquad \text{A7}$$

A.5 Structure of the Potential Matrix V(r)

The components of $V(r)$ are functionals of the background solution $(\bar{A}(r), \bar{B}(r), \bar{\phi}(r))$and its derivatives. Their physical interpretation is as follows:

$V_{gg}(r)$: The **geometric potential**. This term arises from the variation of the Einstein-Hilbert action and quantifies the resistance of the flexible spacetime geometry to deformations. In the bound state, it is positive-definite and provides a restorative potential barrier, embodying the elasticity of the confined spatial geometry. Its specific form is modified by the presence of the background scalar field $\bar{\phi}$. It is positive definite in the bound state and provides a barrier against deformations, representing the stiffness of spacetime.

$V_{\phi\phi}(r)$: The **effective potential for the field perturbation**. It contains several contributions:

$$V_{\phi\phi}(r) = \mu_{eff}^2(r) + \frac{l(l+d-1)}{r^2}\bar{A}(r) + terms\ involving\ \partial_r\bar{\phi},$$

where $\mu_{eff}^2(r)$ may be locally modified from the bare $-{}^2$ by the background curvature. The term proportional to $l(l+d-1)/r^2$ provides a positive centrifugal barrier.

$V_{g\phi}(r)=V_{\phi g}(r)$: The **geometry-matter coupling potential**. This is the crucial off-diagonal term responsible for stabilization. It is proportional to the strength of the background field gradient:

$$V_{g\phi}(r) \propto \sqrt{\kappa_d}\,\partial_r\bar{\phi}(r)\cdot f\left[\bar{g}_{\mu\nu}(r)\right]$$

Since $\bar{\phi}(r)$ varies from a central maximum to zero at the boundary $r=R$, its gradient is non-zero throughout the interior, making this coupling term significant everywhere, especially near the boundary layer.

A.6 Stability Criterion from the Eigenvalue Problem

The system described by Eq.A6 is dynamically stable if and only if the operator on the left-hand side has no negative eigenvalues ω^2. That is, stability requires $\omega^2>0$ for all perturbation modes, ensuring an oscillatory response $e^{-i\omega t}$ with real ω.

The off-diagonal coupling terms $V_{g\phi}$ hybridize the primitive "geometry mode" and "field mode." In the absence of coupling ($V_{g\phi}=0$), the field channel with potential $V_{\phi\phi}$ could possess negative eigenvalues due to the dominant $-\mu^2$ term, leading to instability. However, the non-zero coupling $V_{g\phi}$ mixes this mode with the stable geometric mode governed by the positive potential V_{gg}. This hybridization, analogous to an avoided crossing in a two-level quantum system, pushes the eigenvalues of the resulting mixed modes upward. For a sufficiently strong background gradient $\partial_r \bar{\phi}$ —which is guaranteed by the confinement condition $\bar{\phi}(R) = 0$—the lowest eigenvalue ω^2 becomes positive.

Therefore, solving the eigenvalue problem (A6) with the background from Chapter II confirms that all perturbation modes have $\omega^2>0$. The lowest non-trivial mode ($l=2$) corresponds to a stable quadrupolar (ellipsoidal) oscillation of the bound state. This mathematically substantiates the physical picture that the flexible geometry provides a restorative feedback mechanism that completely quenches the tachyonic instability.

Appendix B: General Variational Principle, Extended Field Equations, and the Zero-Energy Bound State as a Special Case

This appendix presents a first-principles derivation of the coupled geometry-matter system from a constrained action extremum principle. It generalizes the framework of Appendices A and B, demonstrating that the zero-energy bound state emerges as a special, self-consistent solution within a broader class of theories allowing for non-minimal coupling and open systems. The goal is to provide a foundational theoretical basis from which the specific phenomenology discussed in the main text and previous appendices can be derived.

B.1 The Constrained Action Extremum Principle

The fundamental postulate is that the physical evolution of the coupled system—described by the spacetime metric $g_{\mu\nu}$ and the tachyonic scalar field ϕ—is determined by a constrained extremization of a total action.

1. **Total Action:** We consider a general action functional defined over a -dimensional spacetime manifold M:

$$S_{total}[g_{\mu\nu}, \phi, \lambda] = \int_M d^{d+1}x\sqrt{-g}\,L_{total} + S_{boundary} + S_{constraint}$$

The total Lagrangian density is decomposed as:

$$L_{total} = L_G + L_m + L_{int}$$

$L_G = \frac{1}{2\kappa}R$ is the gravitational Einstein-Hilbert Lagrangian, where $\kappa=8\pi G_{d+1}$ and R is the Ricci scalar. $L_m = \frac{1}{2}g^{\mu\nu}\nabla_\mu\phi\nabla_\nu\phi - V(\phi)$ is the matter Lagrangian for the tachyonic field with potential $V(\phi) = -\frac{1}{2}\mu^2\phi^2$. For our university, $V(\phi) = \frac{1}{2}m^2\phi^2$ for $m^2 > 0$.

Lintis the **interaction Lagrangian** encoding direct, non-minimal couplings between ϕ and curvature invariants (e.g., $f(\phi)R$, $h(\phi)R_{\mu\nu}\nabla^\mu\phi\nabla^\nu\phi$). This term is the generalization beyond standard General Relativity. In the simplest case of minimal coupling, $L_{int}=0$.

$S_{boundary}$ is the Gibbons-Hawking-York boundary term and any necessary matter boundary terms required for a well-posed variational principle.

$S_{constraint} = \int_M d^{d+1}x\sqrt{-g}\lambda C[g_{\mu\nu}, \phi]$ implements a physical constraint via the Lagrange multiplier field $\lambda(x)$. The constraint functional $C[g,\phi]=0$ embodies a key physical condition on the system.

2. **The Variational Principle:** The physical configurations ($g_{\mu\nu}(x),\phi(x)$) are those that satisfy $\delta S_{total}=0$ for independent variations of the metric $\delta g_{\mu\nu}$, the field $\delta\phi$, and the Lagrange multiplier $\delta\lambda$. Variation with respect to λsimply enforces the constraint $C=0$.

B.2 Derivation of the Generalized Field Equations

Performing the variations yields the coupled field equations.

1. **Generalized Einstein Equations:** Varying with respect to the inverse metric $g_{\mu\nu}$ gives:

$$\frac{1}{\sqrt{-g}}\frac{\delta\left(\sqrt{-g}L_{total}\right)}{\delta g^{\mu\nu}} + \lambda\frac{\delta C}{\delta g^{\mu\nu}} = 0$$

After standard computation (and assuming Lintis a function of ϕ, R, and possibly $R_{\mu\nu}$), this takes the form of a modified Einstein equation:

$$G_{\mu\nu} = \kappa\left(T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(int)}\right) + \kappa\lambda T_{\mu\nu}^{(C)} \quad \text{(B.1)}$$

Here, $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$ is the Einstein tensor.

$T_{\mu\nu}^{(m)} = -\frac{2}{\sqrt{-g}}\frac{\delta(\sqrt{-g})L_m}{\delta g^{\mu\nu}} = \nabla_\mu\phi\nabla_\nu\phi - g_{\mu\nu}\left[\frac{1}{2}(\nabla\phi)^2 + V(\phi)\right]$is the canonical stress-energy tensor of the ϕ field. $V(\phi) = -\frac{1}{2}\mu^2\phi^2$ or $V(\phi) = \frac{1}{2}m^2\phi^2$.

$T_{\mu\nu}^{(int)} = -\frac{2}{\sqrt{-g}}\frac{\delta(\sqrt{-g})L_{int}}{\delta g^{\mu\nu}}$ is the stress-energy contribution from the non-minimal interaction term. Its specific form depends on the choice of L_{int}.

$T_{\mu\nu}^{(C)} = -\frac{2}{\sqrt{-g}}\frac{\delta(\sqrt{-g})C}{\delta g^{\mu\nu}}$ is the geometric contribution arising from the constraint.

2. **Generalized Tachyon/ Field Equation:** Varying with respect to ϕ gives:

$$\frac{2}{\sqrt{-g}}\frac{\delta\left(\sqrt{-g}\right)L_{total}}{\delta\phi} + \lambda\frac{\delta C}{\delta\phi} = 0$$

This yields a modified Klein-Gordon equation:

$$\Box\phi - \mu^2\phi + \frac{\partial L_{int}}{\partial\emptyset} - \nabla_\mu\left(\frac{\partial L_{int}}{\partial(\nabla_\mu\phi)}\right) + \lambda\frac{\delta C}{\delta\phi} = 0 \quad \text{(B.2a)}$$

$$\Box\phi + m^2\phi + \frac{\partial L_{int}}{\partial\emptyset} - \nabla_\mu\left(\frac{\partial L_{int}}{\partial(\nabla_\mu\phi)}\right) + \lambda\frac{\delta C}{\delta\phi} = 0 \quad \text{(B.2b)}$$

The terms involving Lintand C represent explicit forces on the field due to geometry and the system constraint, beyond the minimal coupling contained in the covariant d'Alembertian □.

B.3 The Zero-Energy Bound State as a Critical Solution

The specific physical scenario of a self-contained, stable bound state is recovered by imposing a particular **energy closure constraint**.

1. **The Global Energy Constraint:** For a static, spherically symmetric configuration, we demand the system's total gravitating mass-energy (e.g., the ADM mass or the quasi-local mass within a radius R) vanishes. This can be encoded in a constraint functional. A convenient choice for a ball of coordinate radius R is:

$$C[g,\phi] = \int_0^R dr \int d\Omega_{d-1}\left(NH + N^iH_i\right) - 0$$

where H and H_i are the Hamiltonian and momentum constraints of General Relativity, and N, N^i are the lapse and shift. In the static gauge and for spherical symmetry, this essentially imposes that the integral of the energy density over the ball, computed from the full stress-energy tensor (including gravitational binding energy), is zero. For the minimal coupling case ($L_{int}=0$), this reduces precisely to the condition $E_{total}=0$ used in Appendix A.2.

2. **Recovery of the Appendix A Solution:** Impose the constraint $C=0$ via the Lagrange multiplier λ. Assume minimal coupling ($L_{int}=0$) and a static, spherically symmetric ansatz for both metric and field. The field equation (B.2) reduces to the Helmholtz-type equation $\nabla^2\phi=-\mu^2\phi$ within the ball. The Einstein equations (B.1), with the energy constraint, determine the background metric. In the weak-field limit (or for a test field in a fixed Minkowski background), the metric becomes approximately flat, and the system reduces to the one solved in Appendix A. The boundary condition $\phi(R)=0$ emerges not as an ad hoc requirement but as a **consequence of the field equation and the energy constraint** applied at a finite boundary. The static, regular solution is again:

$$\phi_0(r) = \phi_0 \frac{J_{(d-2)/2}(\mu r)}{J_{(d-2)/2}(\mu R)} \;\; , \;\text{with}\; \mathrm{J}_{(\mathrm{d}-2)/2}(\mu\mathrm{R}) = 0$$

Or $$\phi_0(r) = \phi_0 \frac{J_{(d-2)/2}(mr)}{J_{(d-2)/2}(mR)} \;\; , \;\text{with}\; \mathrm{J}_{(\mathrm{d}-2)/2}(\mathrm{mR}) = 0$$

The Lagrange multiplier λ adjusts self-consistently to satisfy the zero-energy condition globally. This demonstrates that the zero-energy bound state is a **critical solution** of the constrained action principle.

3. **The "Spring Constant" and Geometric Feedback:** The constraint $C=0$ plays the role of a global constitutive relation. The Lagrange multiplier λ can be interpreted as a measure of the tension or pressure required on the boundary to maintain the zero-energy condition. In the equation for perturbations, this contributes to the effective potential matrix $V(r)$, providing the restoring force that stabilizes oscillations. The condition $J_{(d-2)/2}(\mu R)=0$ thus appears as a quantization condition for stable equilibrium, linking the field's internal scale (μ^{-1}) to the geometric scale (R).

B.4 Extension to Open Systems and the Energy Spectrum

The framework naturally generalizes to non-zero energy states, describing an "open" or "excited" system.

1. **Relaxing the Constraint:** Instead of $C=0$, we can impose $C=E_0$, where E_0 is a fixed (non-zero) energy. This represents a system in contact with a reservoir that fixes its total energy. The variational principle now seeks an extremum of the action subject to this fixed energy.

2. **The Generalized Static Solution:** Solving equations (B.1) and (B.2) with the constraint $C=E_0$ will yield a one-parameter family of static, spherical solutions parameterized by E_0. The field profile $\phi(r;E_0)$ and the concomitant metric will differ from the zero-energy case. The boundary condition at $r=R$ will generally be more complex than a simple Dirichlet condition, potentially involving a non-zero field derivative, reflecting a balance between internal field stress and boundary pressure/tension.

3. **Energy Spectrum and Stability:** As E_0 varies, we obtain a spectrum of equilibrium configurations. The zero-energy solution ($E_0=0$) is expected to be the ground state of this spectrum. Solutions with $E_0>0$ (positive total energy) or $E_0<0$ (negative total energy relative to infinity) are excited states. The linear stability analysis, when applied to these generalized solutions using the equations derived from (B.1) and (B.2), will determine their stability. The zero-energy ground state is anticipated to be stable, while excited states may possess unstable modes, corresponding to decay to the ground state via radiation or other processes.

4. **Coupling Constant and Observables:** The interaction Lagrangian L_{int}, if present, introduces new dimensionless coupling constants (e.g., $\xi in\ \xi\phi^2 R$). These constants modify the relation between μ, R, and the energy E_0. They also affect the predicted vibration frequencies ω (from perturbation analysis) and the detailed anisotropic signatures in experimental probes. Thus, the general theory is falsifiable and predictive.

B.5 Radial Reduction under Static Spherical Symmetry

To obtain explicit static, spherically symmetric bound-state solutions, we reduce the general field equations (B.1) and (B.2) under the following physically motivated assumptions:

1. **Physical Setup**:

Consider a real massive scalar field with potential $V(\phi)=+\frac{1}{2}m^2\phi^2$.

Assume minimal coupling, i.e., $L_{int}=0$.

The global zero-energy constraint is enforced via boundary conditions.

2. **Symmetry Ansatz**:

The metric is static and spherically symmetric:

$$ds^2 = -A(r)dt^2 + B(r)dr^2 + r^2 d\Omega_{d-1}^2$$

where $d\Omega_{d-1}^2$ is the standard metric on the (d−1)-sphere.

The scalar field depends only on the radial coordinate: $\phi=\phi(r)$.

3. **Reduced Field Equations**:

Scalar Field Equation: Equation (B.2) reduces to

$$\frac{1}{\sqrt{AB}r^{d-1}}\frac{d}{dr}\left(\sqrt{\frac{A}{B}}r^{d-1}\frac{d\phi}{dr}\right) + m^2\phi = 0 \tag{B3}$$

Einstein Equations: It is convenient to introduce a "mass function" M(r)defined by

$$B(r) = \left[1 - \frac{2\kappa_d M(r)}{(d-2)r^{d-3}}\right]^{-1}$$

where $\kappa_d=8\pi G_{d+1}$. The *tt* and *rr* components of the Einstein equations (C.1) then yield, respectively:

$$\frac{dM}{dr} = \frac{\Omega_{d-1}}{2} r^{d-2}\left[\frac{1}{B}\left(\frac{d\phi}{dr}\right)^2 + m^2\phi^2\right] \tag{B4}$$

$$\frac{A'}{A} = \frac{2\kappa_d M(r) + \kappa_d r^{d-1}\left[\frac{1}{B}\left(\frac{d\phi}{dr}\right)^2 - m^2\phi^2\right]}{(d-2)r^{d-3}\left[1 - \frac{2\kappa_d M(r)}{(d-2)r^{d-3}}\right]} \tag{B5}$$

where $\Omega_{d-1}$is the area of the unit (d−1)-sphere and a prime denotes a derivative with respect to r.

4. **Boundary Conditions**:

Regularity at the Origin: As $r\to 0$, we require $\phi(0)$ to be finite, $\phi'(0)=0$, $M(0)=0$, and $A'(0)=0$.

Boundary of the Bound State: At a finite radius r=R, we impose the zero-energy condition and the vanishing of the field:

$$M(R)=0,\ \phi(R)=0. \tag{B6}$$

Equations (B.4)–(B.10) form a set of coupled ordinary differential equations for the functions $\phi(r)$, M(r), and A(r). They constitute the foundation for numerically solving for the static, spherically symmetric background solution of the bound state.

B.6 Conclusion: A Unified Framework

This appendix has elevated the specific model of Appendices A and B into a comprehensive theoretical framework based on a constrained action principle. The key results are:

1. The coupled geometry-matter dynamics, including potential beyond-GR interactions, derive from the extremization of a general action subject to a physical constraint (e.g., on total energy).

2. The remarkable zero-energy tachyonic bound state—with its Dirichlet boundary condition, quantized radius, and proven stability—emerges as the unique ground state solution when the constraint imposes zero total mass-energy.

3. The framework naturally extends to open systems with non-zero energy, yielding a spectrum of equilibrium configurations whose stability can be systematically analyzed.

4. The geometric feedback mechanism, mathematically expressed through the constraint and the coupling in the field equations, is identified as the fundamental origin of stability, turning the tachyonic instability into a spectrum of stable vibrational modes.

Therefore, the specific results presented in the main text and in Appendices A and B are not isolated phenomena but are the foundational predictions of a broader, self-consistent theoretical structure. This structure provides a principled basis for further exploration, including the detailed study of excited states, the inclusion of additional fields, and the computation of precise observational signatures for experimental tests.

Appendix C: Spacetime-Matter Topological Duality and the Interaction Dynamical Mechanism

C.1 Introduction: The Unity of Spacetime and Matter

Classical physics often decomposes physical systems into two independent parts: the "background spacetime" and the "dynamic matter." However, from a deeper perspective, spacetime geometry and matter fields constitute an inseparable dual system. The theoretical framework presented here indicates that any physical system must simultaneously satisfy two distinct types of constraints: continuous geometric deformation (**longitudinal**) and discrete topological structure (**angular**). These two sets of degrees of freedom are not independent but are interwoven, collectively determining the evolution of the universe.

C.2 Decomposition of Degrees of Freedom: Longitudinal Flexibility and Angular Rigidity

To achieve self-consistent coexistence, the total action of a physical system must incorporate the dual attributes of geometry and matter. We decompose the system's degrees of freedom into two orthogonal dimensions:

Longitudinal Dimension (Longitudinal/Flexible): Continuous Geometric Response

Physical Image: Imagine spacetime as an elastic membrane, its scale described by a radius R or the metric $g_{\mu\nu}$.

Mathematical Property: Continuity. The expansion/contraction of space is smooth, able to change continuously from R_0 to R_0+dR. This corresponds to the spacetime curvature described by General Relativity.

Dynamical Role: When the energy density of internal matter changes, space responds through continuous expansion or contraction. This regulatory mechanism is very "cheap" as it does not involve changes in topological structure. On cosmological scales, this means space can adapt to minute energy fluctuations by changing its radius R, manifesting as the expansion or contraction of the universe.

Angular/Toroidal Dimension (Angular/Rigid): Discrete Topological Constraints

Physical Image: Matter moves on this elastic membrane, or space itself possesses a winding structure (e.g., cosmic strings, spin networks).

Mathematical Property: Discreteness. Angular motion is strictly protected by topology. One cannot have "half a winding" or "1.5 loops." This directly leads to the emergence of quantization conditions—angular momentum $L=n\hbar,$ magnetic quantum numbers m must be integers.

Dynamical Role: Angular degrees of freedom cannot be adapted via fine-tuning. To change a system's angular quantum number, a discontinuous quantum jump (energy absorption/emission) must occur. On cosmological scales, this means imparting overall rotation to the entire universe requires overcoming enormous centrifugal forces and altering its topological structure, which is energetically extremely costly.

C.3 Symmetry Breaking and Cosmological Observations

Applying the above principles to our observed universe leads to a striking conclusion: on cosmological scales, longitudinal degrees of freedom are "cheap" and active, while angular degrees of freedom are "expensive" and frozen.

Activity of the Longitudinal: Space is extremely flexible longitudinally. Therefore, when the total energy of the universe experiences minor fluctuations, the system's preferred regulatory method is to change the radius R. This is the observed expansion of the universe—a low-energy, continuous mode of regulation.

Freezing of the Angular: To induce overall rotation in a universe with a radius $R \sim 10^{26} m$ would require energy far exceeding the total energy of the universe. Furthermore, rotation would generate intense centrifugal forces, disrupting the isotropy of the Cosmic Microwave Background (CMB) radiation. Consequently, under the principle of minimum energy, angular degrees of freedom are "frozen" in the lowest energy state, $n=0$ (no rotation). The absence of observed macroscopic rotation is not due to a lack of angular momentum, but because the cost of activating angular degrees of freedom is prohibitive.

Within the unified framework, fundamental interactions differ in how their range couples to the global geometry of the universe. The photon, as a massless gauge boson, mediates the long-range electromagnetic force. Its propagation and energy (frequency) are directly modulated by the dynamics of the cosmic scale factor $R(t)$, as manifested in cosmological redshift; thus, it is strongly coupled to the longitudinal stretching of the background spacetime. In contrast, the weak force mediated by the $W^{\pm}$ and Z bosons—which acquire mass via the Higgs mechanism—and the strong force (mediated by gluons), which is short-ranged due to color confinement, are all short-range forces. They possess an intrinsic mass scale or confinement energy scale, restricting their interactions to microscopic, local domains. Consequently, they are largely decoupled from the cosmological evolution of $R(t)$. This distinction fundamentally corresponds to the continuous, global longitudinal degrees of freedom versus the discrete, local angular degrees of freedom.

C.4 Decoupling Mechanism of Interaction

The coupling between spacetime and matter is not mediated by a specific "interaction force" but is realized through a global zero-energy constraint. Mathematically, this constraint manifests as a topological boundary condition, requiring the system to balance longitudinal deformation and angular excitation.

Weak Coupling Limit (Everyday and Cosmological Scales):

When the energy density of the matter field ϕ is low, or spacetime curvature varies gently, the interaction between longitudinal and angular components is weak. The system can then decouple into two independent equations:

The geometric part reduces to the Einstein Field Equations, describing how spacetime minimizes energy via continuous scaling.

The matter part reduces to the Klein-Gordon or Dirac equations, describing quantum behavior of particles on a fixed background.

Strong Coupling Limit (Black Holes, Tachyon):

In extreme environments with extremely high matter density or curvature (e.g., black hole interiors,), the boundary between longitudinal and angular degrees of freedom dissolves, and the system enters a highly correlated, strong-coupling state. Here, the classical descriptions of matter and geometry fail; the spacetime background undergoes violent fluctuations, and the topological nature of the matter field becomes fundamentally intertwined with the flexible deformations of geometry.

In this regime, the equations describing the system's evolution reduce to the Tachyon Sea Model proposed in this paper. In this model, the dominant matter field consists of the Tachyon Field with imaginary mass. The core dynamics of the model are described by a set of non-linear evolution equations coupling the imaginary-mass tachyon field ϕ and the dimensional radius R of space (or the "shell"). In this model, the co-evolution of the imaginary-mass tachyon field and the spatial dimensional radius jointly depicts a non-singular physical picture of the high-energy state of the early universe or the core region of a black hole.

C.5 **Zero-Energy Hypothesis for Construct duality constraint equation** $C[\phi, g]$

As previously elaborated, the physical laws of our universe are the product of mutual compromise and co-evolution between spacetime and matter under a global zero-energy condition. To mathematically describe this "compromise" mechanism from first principles, we construct the duality constraint equation $C[\phi,g]=0$. The formulation of this equation is guided by the following four core postulates. Together, they define the mathematical and physical structure of the Zero-Energy Principle for flexible spacetime, aiming to describe spacetime dynamics and fundamental interactions within a unified geometric logic. A key aspect of this logic is that the internal gauge symmetries of matter fields (such as U(1), SU(2), SU(3)) are understood as being projected onto or inherently constrained by the fundamental SO(3) rotational symmetry of the spatial geometry of our universe.

1a）Closed Space（Tachyon-space state or entire cosmos）: Zero-Energy Postulate: The total Hamiltonian of the system strictly vanishes, $H_{total} = H_{matter} + H_{geometry} = 0$. This serves as the primary constraint for all physical processes.

1b）Open Space（matter-space state）: Minimum Geometric Energy Postulate: In an open spatial domain, the geometry evolves to minimize its positive energy contribution subject to the zero-energy constraint ($\delta \int L_{geo} d^4 x = 0$). This determines the "most economical" way for spacetime to curve in response to and in adaptation with the matter distribution, defining the specific stable configurations (like bound states) that arise from the interplay.

2）Angular Rotational Closure Postulate: The system maintains zero-energy invariance under SO(3) rotations ($[J_i, H]_{total} = 0$). We propose that spin emerges naturally as the geometric degree of freedom required to preserve this symmetry.

3）Longitudinal general Covariance Postulate: The fundamental equations are generally covariant, ensuring that the zero-energy constraint holds true across all coordinate transformations (i.e., *Lorentz invariance*).

C6 Exemplary Constructions of the Adaptation Functional C[Φ,g]

Field Type	**Exemplary Construction of the Adaptation Functional C[Φ,g]& Term-by-Term Explanation**
1. Massive Field (Scalar field ϕ): $C[\phi, g] = \int d^4x\sqrt{-g}[\xi R\phi^2 + \lambda(\phi^2 - v^2)^2]$	**Term 1:** $\xi R\phi^2$**:** *Non-minimal coupling term.* Establishes a direct, local interaction between the scalar field amplitude ϕ^2 and the spacetime curvature scalar *R*. This term forces the geometry to adapt to the field's magnitude and vice-versa, with coupling strength ξ. **Term 2:** $\lambda(\phi^2 - v^2)^2$**:** *Self-interaction potential*(e.g., Higgs potential). It defines the field's intrinsic tendency to stabilize around a vacuum expectation value v, representing the matter field's own stability condition within the adaptable geometry.
2. Electromagnetic Field (U(1) Gauge field A_μ):	**Term 1:** $\left(\oint_{\partial M} A\right)^2$**:** *Boundary holonomy constraint* .Squares the circulation of the

$C[A,g] = \left(\oint_{\partial M} A\right)^2 + \eta \int_M F \wedge F$	gauge potential A along the spatial boundary ∂M. This term imposes a global, topological condition related to magnetic flux, explicitly enforcing a closure constraint on the U(1) field. **Term 2:** $\eta \int_M F \wedge F$**:** *Topological term.* The integral of $F \wedge F$ is a topological invariant (e.g., related to the instanton number on a closed manifold). This term encodes the adaptation of the electromagnetic field's global topological structure to the spacetime geometry.
3. Electroweak Field ($SU(2)_L \times U(1)_Y$ + Higgs Φ): $C[W,B,\Phi,g] = \int d^4x \sqrt{-g} Tr\left[\left(D_\mu \Phi\right)^\dagger \left(D^\mu \Phi\right)\right] f(R) + \kappa\left(\int Tr(W \wedge W)\right)^2$	**Term 1:** $\sqrt{-g} Tr\left[\left(D_\mu \Phi\right)^\dagger \left(D^\mu \Phi\right)\right] f(R)$**:** *Curvature-coupled Higgs kinetic term.* Couples the kinetic energy of the Higgs field (governing electroweak symmetry breaking) to a function of the Ricci scalar *f(R)*. This creates a dynamic interplay where the process of symmetry breaking and local geometry co-adapt. **Term 2:** $\kappa\left(\int Tr(W \wedge W)\right)^2$**:** *Global non-Abelian topological constraint.* The integral ∫Tr(W∧W) corresponds to the second Chern number of the SU(2) gauge field. Squaring it constrains this topological charge, enforcing a strong closure condition for the non-Abelian degrees of freedom.
4. Strong Force ($SU(3)_c$ Gauge field $G^a{}_\mu$): $C[G,g] = \int d^4x \sqrt{-g}\theta Tr\left(G \wedge \tilde{G}\right) + \zeta\left(\int d^4x \sqrt{-g} Tr\left(G_{\mu\nu} G^{\mu\nu}\right)\right)^2$	**Term 1:** $\sqrt{-g}\theta Tr\left(G \wedge \tilde{G}\right)$**:** The QCD θ-term. This topological density term couples the gluon field's topological configuration (instantons) directly to the spacetime volume element. It represents a fundamental adaptation between the topology of the strong force and the geometry. **Term 2:** $\zeta\left(\int d^4x \sqrt{-g} Tr\left(G_{\mu\nu} G^{\mu\nu}\right)\right)^2$**:** Global constraint on the Yang-Mills action. Squares the total Yang-Mills action of the gluon field. This term forces an extremization of the overall field energy in adaptation with geometry, acting as a global energy-scale matching condition.

C.7 : Bound State of a Positive-Mass Scalar Field in Flexible Geometry – A Bridging Example

To demonstrate the generalization from the tachyon bound state (where zero-energy holds strictly) to generic open systems (minimum geometric energy), we consider the simplest case of a positive-mass scalar field ϕ (potential $V(\phi)=\frac{1}{2}m^2\phi^2$, $m^2>0$) in a static, spherically symmetric metric. The system is no longer a strictly zero-energy closed system, but an open "matter–geometry" adaptive configuration governed by the Extremum-Adaptation Principle (Section IV.3.2).

C7.1 Setup and Equations

Take the total action (Eq. (8)) with $L_m=-\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi-\frac{1}{2}m^2\phi^2$, and adopt the simplest form of the adaptation functional for a scalar field from Appendix C6:

$$C[\phi,g]=\xi R\phi^2+\lambda(\phi^2-v^2)^2,$$

where ξ is the non-minimal coupling constant, λ the self-interaction strength, and v the vacuum expectation value (can be set to zero). For simplicity, let $v=0$, $\lambda=0$, keeping only the non-minimal coupling term $\xi R\phi^2$. Then the variational equations (9)–(10) become:

$$G_{\mu\nu}=8\pi G\left[T^{(m)}_{\mu\nu}+\beta\frac{\delta C}{\delta g^{\mu\nu}}\right],$$

$$\Box\phi-m^2\phi=-\beta\frac{\delta C}{\delta\phi}.$$

Computing $\delta C/\delta g^{\mu\nu}$ and $\delta C/\delta\phi$ (using $\delta R=R_{\mu\nu}\delta g^{\mu\nu}+g_{\mu\nu}\Box\delta g^{\mu\nu}-\nabla_\mu\nabla_\nu\delta g^{\mu\nu}$) yields:

$$\frac{\delta C}{\delta g^{\mu\nu}}=\xi(\phi^2G_{\mu\nu}+g_{\mu\nu}\Box\phi^2-\nabla_\mu\nabla_\nu\phi^2),\frac{\delta C}{\delta\phi}=2\xi R\phi.$$

Substituting these gives the coupled system:

$$G_{\mu\nu}=8\pi G\left[T^{(m)}_{\mu\nu}+\beta\xi\left(\phi^2G_{\mu\nu}+g_{\mu\nu}\phi^2-\nabla_\mu\nabla_\nu\phi^2\right)\right] \qquad \text{C1}$$

$$\phi-m^2\phi=-2\beta\xi R\phi \qquad C2$$

C7.2 Static Spherically Symmetric Reduction

Assume the metric takes the form $ds^2=-A(r)dt^2+B(r)dr^2+r^2d\Omega^2$, and the field $\phi=\phi(r)$. Define $B(r)=[1-2GM(r)/r]^{-1}$(with c=1), and introduce the dimensionless variable $x=mr$. In the weak-field approximation ($GM/r\ll 1$) and neglecting the backreaction of the metric on the field equation (i.e., taking $A=B=1$ as the zeroth-order approximation), Eq. (C2) reduces to:

$$\frac{1}{x^2}\frac{d}{dx}\left(x^2\frac{d\phi}{dx}\right)-\phi=-2\beta\xi R\phi.$$

Here R is the background curvature scalar. In the flat approximation $R=0$, this becomes the standard Klein–Gordon equation, whose solution decays exponentially at infinity and cannot form a bound state of finite extent. However, once metric backreaction is included (so that $R\neq 0$), the equation becomes:

$$\nabla^2\phi=(m^2-2\beta\xi R)\phi.$$

If $2\beta\xi R>m^2$, the effective mass squared becomes negative, giving the field an effective "tachyonic" behavior, which may allow the formation of a standing-wave bound state within a finite region. This is the core mechanism by which flexible geometry (through curvature R) regulates the field.

C7.3 Numerical Example

Choose parameters $\beta\xi=1$(dimensionless), and assume that inside the bound state the curvature is approximately constant, $R=R_0$. Define the effective mass $m^2_{eff}=m^2-2\beta\xi R_0$. Require $m^2_{eff}<$

0, i.e., $R_0 > m^2/(2\beta\xi)$. The equation then becomes:

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\phi}{dr}\right) + \left|m_{eff}\right|^2\phi = 0,$$

whose regular spherically symmetric solution is $\phi(r)=\phi_0 sin(|m_{eff}|r)/r$. Imposing the Dirichlet boundary condition $\phi(R)=0$ gives $|m_{eff}|R=n\pi$ (na positive integer). Hence the bound-state radius is quantized:

$$R_n = \frac{n\pi}{\sqrt{2\beta\xi R_0 - m^2}}. \qquad \text{C3}$$

This expression is formally identical to the quantization condition for the tachyon bound state, $J_{1/2}(\mu R)=0$ (i.e., $\mu R=n\pi$). The difference is that here the effective mass originates from curvature feedback rather than from a bare imaginary mass. **Thus, in flexible geometry, a positive-mass field can acquire an effective tachyonic behavior through adaptive curvature, thereby forming a bound state.** When $\beta\to 0$, $R_n\to\infty$ and the bound state disappears, recovering the free field.

C7.4 Relation to the Tachyon Bound State

If we further set the bare mass m=0 and let R_0 be determined self-consistently by the field's own curvature (i.e., solving the full system), the problem reduces to the tachyon bound state of Chapter II. Therefore, the positive-mass scalar bound state can be regarded as a generalization of the tachyon bound state under finite curvature feedback, and the non-minimal coupling term $\xi R\phi^2$ in Appendix C6 is precisely the operator that realizes this generalization.

This example demonstrates that the Extremum-Adaptation Principle (Section IV.3.2) can naturally accommodate not only the tachyon special case but also bound states of positive-energy fields in flexible geometry, thereby building a bridge from closed systems to open systems.

C8. Conclusion

The physical reality we observe is a dynamic balance between spacetime geometry (longitudinal/continuous) and matter fields (angular/discrete) under a global zero-energy constraint. The evolution of the universe is the continuous geometric deformation process of this dual system under the premise of conserving topology. Under conventional energy scales, this balance manifests as the separation of gravity and quantum theory. In extreme conditions such as black holes and the origin of the universe, the system evolves into a new physical picture dominated by strong-coupling mechanisms, described jointly by the tachyon field and dynamic geometric radius.